AI Dispatch Advisor

Aircraft MEL/MMEL Compliance Platform

Transforming regulatory complexity into real-time decisions

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PURPOSE

Empowering Flight Operations with Instant Compliance Intelligence

Flight operations teams — dispatchers, maintenance controllers, and operations coordinators — are routinely required to make high-stakes dispatch decisions under extreme time pressure. When equipment is inoperative, every minute of delay compounds operational cost, crew scheduling risk, and passenger impact. Yet the documents governing these decisions — the Minimum Equipment List (MEL) and Master Minimum Equipment List (MMEL) — span hundreds of dense regulatory pages demanding expert interpretation.

The **AI Dispatch Advisor** is an intelligent, real-time decision-support system purpose-built to address this gap. It interprets MEL/MMEL documentation on demand, evaluates the current aircraft status against applicable rules, and delivers actionable, regulation-grounded dispatch guidance — with complete traceability for every recommendation.

Mission

Provide dispatchers and maintenance controllers with instant, reliable, audit-ready compliance guidance — eliminating ambiguity and reducing the risk of costly or unsafe dispatch decisions.

Core Value

Replace manual documentation search with an AI-powered knowledge engine that surfaces the right MEL/MMEL rule, the right operational condition, and the right action — in seconds, not minutes.

Safety Commitment

Every recommendation is grounded in verified regulatory data, fully traceable to source documentation, and designed to support — not replace — qualified human decision-makers.

SYSTEM OVERVIEW

How the AI Dispatch Advisor Works — End to End

The AI Dispatch Advisor operates as a seamlessly integrated, multi-layer intelligence pipeline — from raw regulatory documentation to a clear, actionable dispatch recommendation delivered in seconds.

At ingestion, MEL and MMEL source documents are transformed into a machine interpretable knowledge base using structured extraction and vector embedding. At query time, the dispatcher provides the aircraft identifier and the nature of the inoperative equipment. The AI evaluation engine retrieves the relevant regulatory rules via semantic search, applies operational condition logic, and synthesizes a grounded, citation-backed recommendation. Every decision step is logged for full regulatory traceability — ensuring the system meets audit requirements from day one.

Input

Aircraft tail, fleet type, inoperative system, route, operational conditions

Processing

Semantic retrieval from MEL/MMEL knowledge base, rule matching, condition evaluation

Output

Dispatch GO / NO-GO / conditional GO with required procedures, limitations, and source citations

INDUSTRY CONTEXT

Understanding MEL & MMEL: The Backbone of Dispatch Compliance

What Is the MMEL?

The **Master Minimum Equipment List (MMEL)** is a regulatory document issued by aviation authorities — such as the FAA or EASA — that defines the maximum list of instruments, equipment, and functions that may be inoperative while still permitting safe aircraft dispatch. The MMEL is aircraft-type specific and establishes the outer boundary of permissible interoperability for a given air frame under defined operational conditions.

What Is the MEL?

The **Minimum Equipment List (MEL)** is an operator-specific document derived from the MMEL. Airlines customize their MEL within MMEL limits to reflect their fleet configuration, route structures, and operational procedures. MEL items include required corrective actions, crew procedures, time limits for repair, and flight operation restrictions — making each item a multi-dimensional compliance decision.

Why It Matters Operationally

MEL compliance is among the most operationally critical and time-sensitive tasks in flight dispatch. A dispatcher assessing an inoperative system must:

- Identify the correct MEL item for the affected system
- Evaluate whether dispatch is permitted under current conditions
- Confirm all required maintenance or operational procedures are completed
- Ensure documentation is complete for regulatory audit purposes
- Communicate limitations clearly to the flight crew

With turnaround windows as short as 30–45 minutes, and MEL documents exceeding 500+ pages, the margin for error is very small. Errors in MEL interpretation can result in regulatory violations, aircraft groundings, safety incidents, and significant financial penalties.

> Aviation safety data consistently identifies MEL misapplication as a contributing factor in operational disruptions and regulatory findings — underscoring the urgent need for smarter, faster compliance tools.

RESPONSIBLE AI & SAFETY

Safety-First by Design: Responsible AI in Aviation Operations

Operating in an FAA/EASA- regulated environment demands a standard of AI responsibility that exceeds conventional enterprise software. The AI Dispatch Advisor is architected around three non-negotiable principles: **grounded outputs**, **full traceability**, and **human authority**.

1

Grounded Outputs Only

The system is architecturally prevented from generating responses that are not anchored to verified MEL/MMEL source documentation. Retrieval-augmented generation (RAG) ensures the language model operates exclusively on retrieved regulatory content. When no applicable rule is found, the system explicitly flags the gap and escalates — it never generates an unverifiable compliance recommendation.

2

Full Decision Traceability

Every dispatch recommendation is accompanied by a complete audit trail: the specific MEL item referenced, the regulatory version in effect, the operational conditions evaluated, and the precise reasoning chain applied. These records are immutably stored in Cosmos DB and retrievable on demand for regulatory audits, incident investigations, and quality assurance reviews — meeting the standards of all major aviation regulatory bodies.

3

Human-in-the-Loop Authority

The AI Dispatch Advisor is a decision-support tool — not a decision-making authority. All recommendations require acknowledgment and authorization by a qualified dispatcher or maintenance controller. The system enforces explicit human sign-off at every compliance decision point, and for complex or ambiguous scenarios, automatically escalates to senior operations personnel. Human authority over dispatch decisions is absolute and non-negotiable.